

(1) ✓ low

RADIX SORT

Ra

- (1) grouped as non-comparison based sorting algo's (bucket sort & Radix)
- (2) Also grouped under distribution sorts.
Bucket / Bin Sort Radix Sort
- Unsorted bucket Sorted bucket
- Scatter $\rightarrow$ Sort $\rightarrow$ gather Scatter $\rightarrow$ gather $\rightarrow$ rep
- Relies on stability.

(3) Sorts based on Radix character
(makes 10 Bins / Buckets) as we deal with decimal Not system.

(4) Sorting is done from LSD then we gather & perform sorting loop until all digits are according to which we have ran a loop) this automatically maintains Stability for us.

(5) time Complexity $O(dn) \approx O(n)$
 d = no. of digits of max int in array better than $O(n \log n)$
 n = size of the array.

(6) limited Space 10 bins / buckets
max no. of nodes in each bucket
= the no. of elements (in worst case if all elements fall in 1 bucket) $O(n)$
 $n = 10$

LSD

RAY

221	453	64	125	8	948	33	121
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max = 948

RAY[I]

0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	1	1	0
	221	453	64	125					
	121		33					8	

Scatter - gather

RAY[I] % 10

for gathering

221	121	453	33	64	125	8	948
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follow FIFO

from smallest
idx to largest

948/10 > 0
94 > 0

(max / exp) > 0 | exp = 100
(RAY[I] / 10) % 10

idx to maintain stability

0	1	2	3	4	5	6	7	8	9
	0						0	0	0
	8	221	33	948	453	64			
		121							
		125							

(max / exp) > 0

948/100
9 > 0 | exp = 100

Scatter & gather

8	221	121	125	33	948	453	64
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(RAY[I] / 100) % 10

0	1	2	3	4	5	6	7	8	9
			0		0	0	0	0	
	8	121	221		453				948
		33	125						
		64	948						

8	33	64	121	125	948	221	453
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(max / exp) > 0

948/1000 > 0

0 ≠ 0

Stops